

Teaching Exam

Subject – Computer Science

Data Handling Using Pandas



Lecture No –3

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Topics to be Covered



Topic

[doc, dloc]

Slicing

matplotlib

bar graph

histogram

line graph

um part

Name.

city.

well,

Marks

1987

column

print (df-city)

```
print(df["wt"]) ✓
```

```
print(df["wt", "city"])
```

```
print(d["wt", "City", "Name"])
```

```
print(df.head(2))
```

Prüfung (d.h. 25% (R2)) → gibt
komplett

```
print(d["city"]["R3"])
```

→ Lucknow

loc, illoc

`print(df.loc[:])` → Complete df print.

`print(df.loc[:, ["Name", "wt"]])`
 AU row Given columns Print

`print(df.loc[:, ["Name": "Marks": 2]])`
 ↓
 Rows

`print(df.loc[["R2": "R4"], ["Name", "wt"]])`
 2 3

`print(df.iloc[[1:4], [0, 2]])`

loc. [start row : end row], [start column, end column]
slicing

print(df.loc["R1"])

print(df.loc["R4", "R2"])

print(df.loc["R4", "R2", "R2", "R1"])

print(df.loc["R2": "R4"])
R2, R3, R4

print(df.loc["R1": : 2])
R1, R3, R5

Start, end, skip
 — : —

...

◆ Why Do We Plot?

Plotting helps:

- Visualize trends in data
- Understand relationships
- Make comparisons easier
- Present findings clearly

univariate data

Month	Sales Amount (₹)
January	1,20,000
February	98,500
March	1,35,000
April	1,10,750
May	1,42,300
June	1,25,600
July	1,30,000
August	1,48,900
September	1,20,750



pip install matplotlib

import pandas as pd.

import matplotlib.pyplot as plt

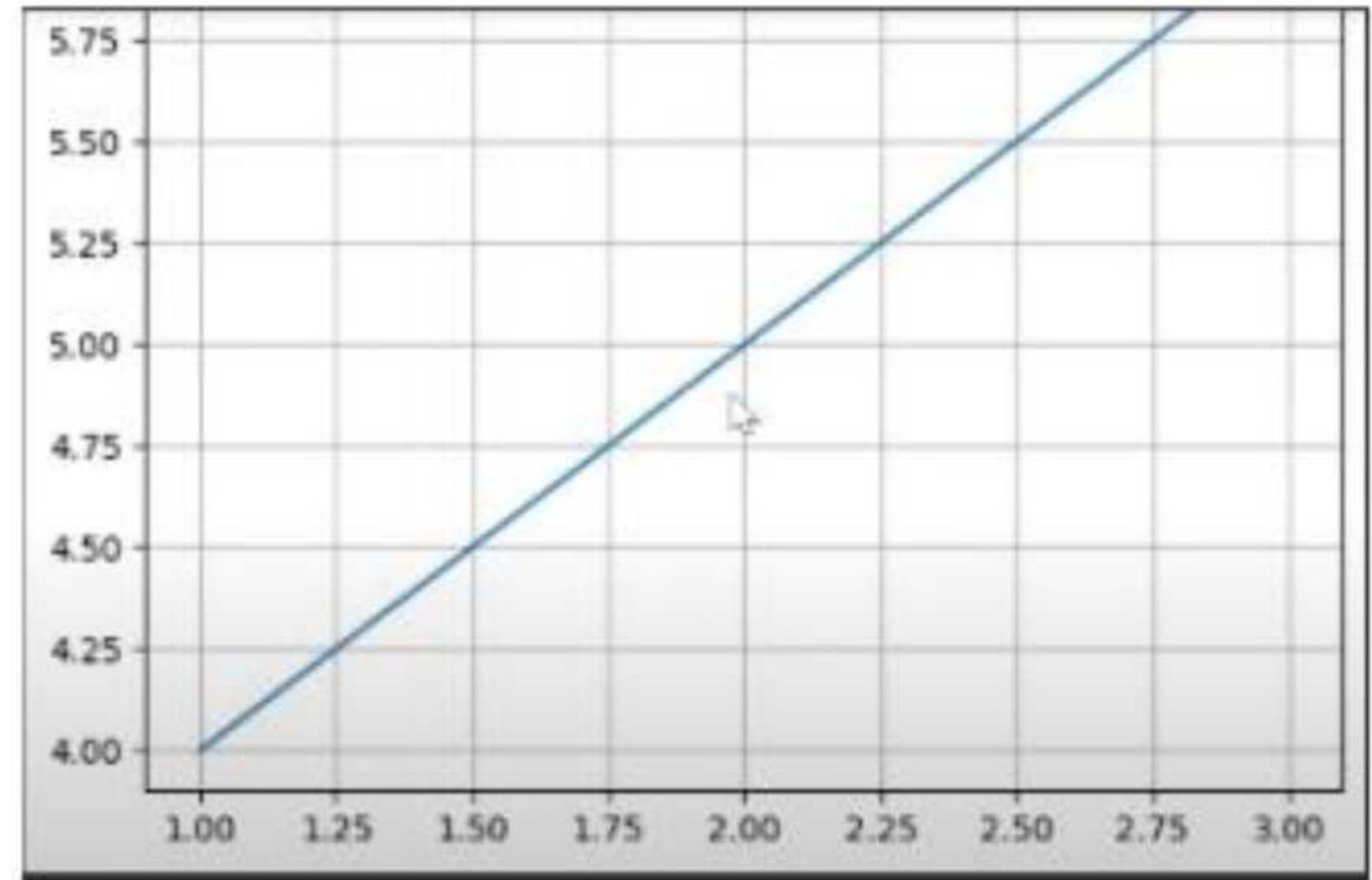
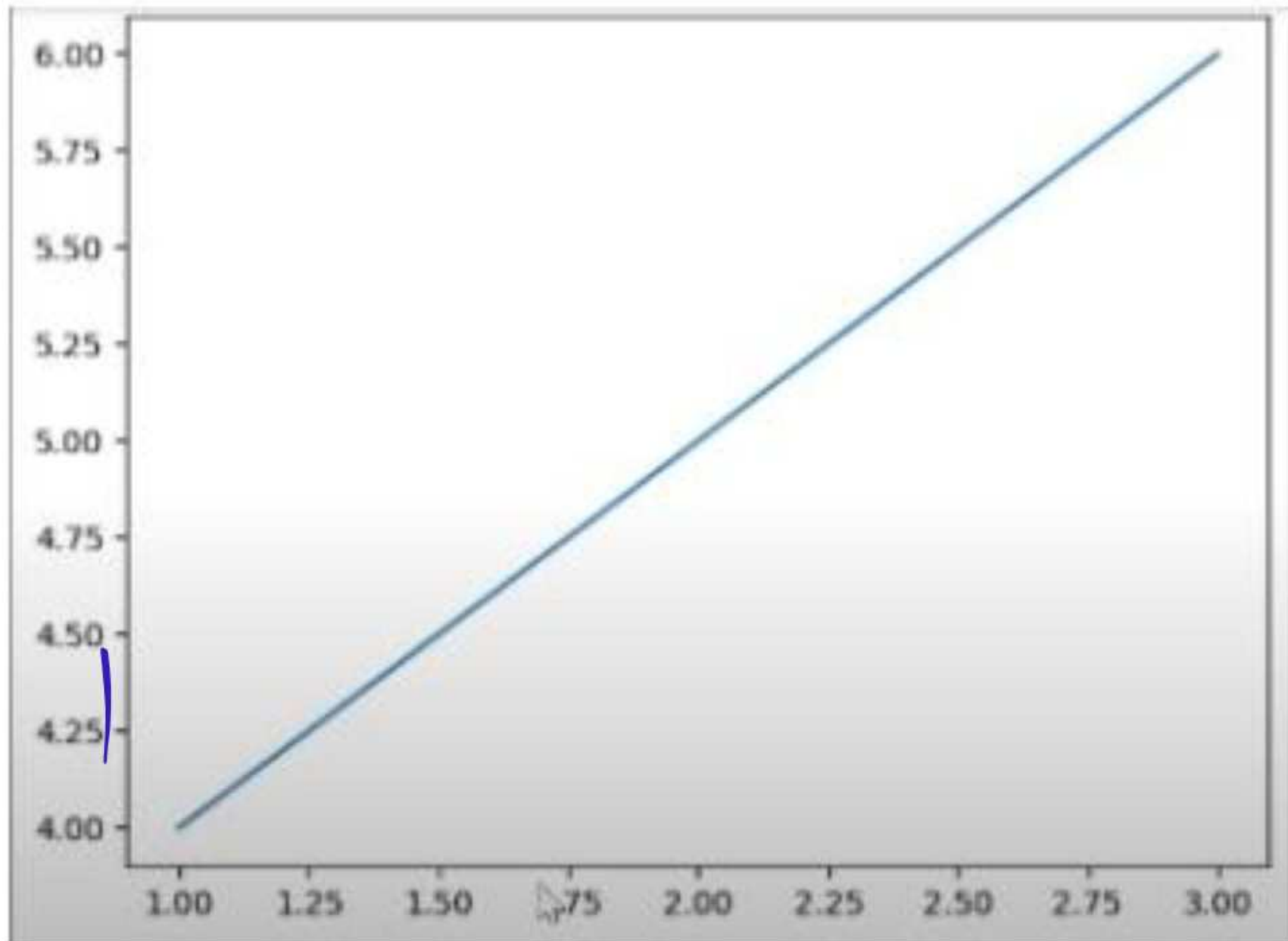
a = [1, 2, 3] # (1,4), (2,5), (3,6)

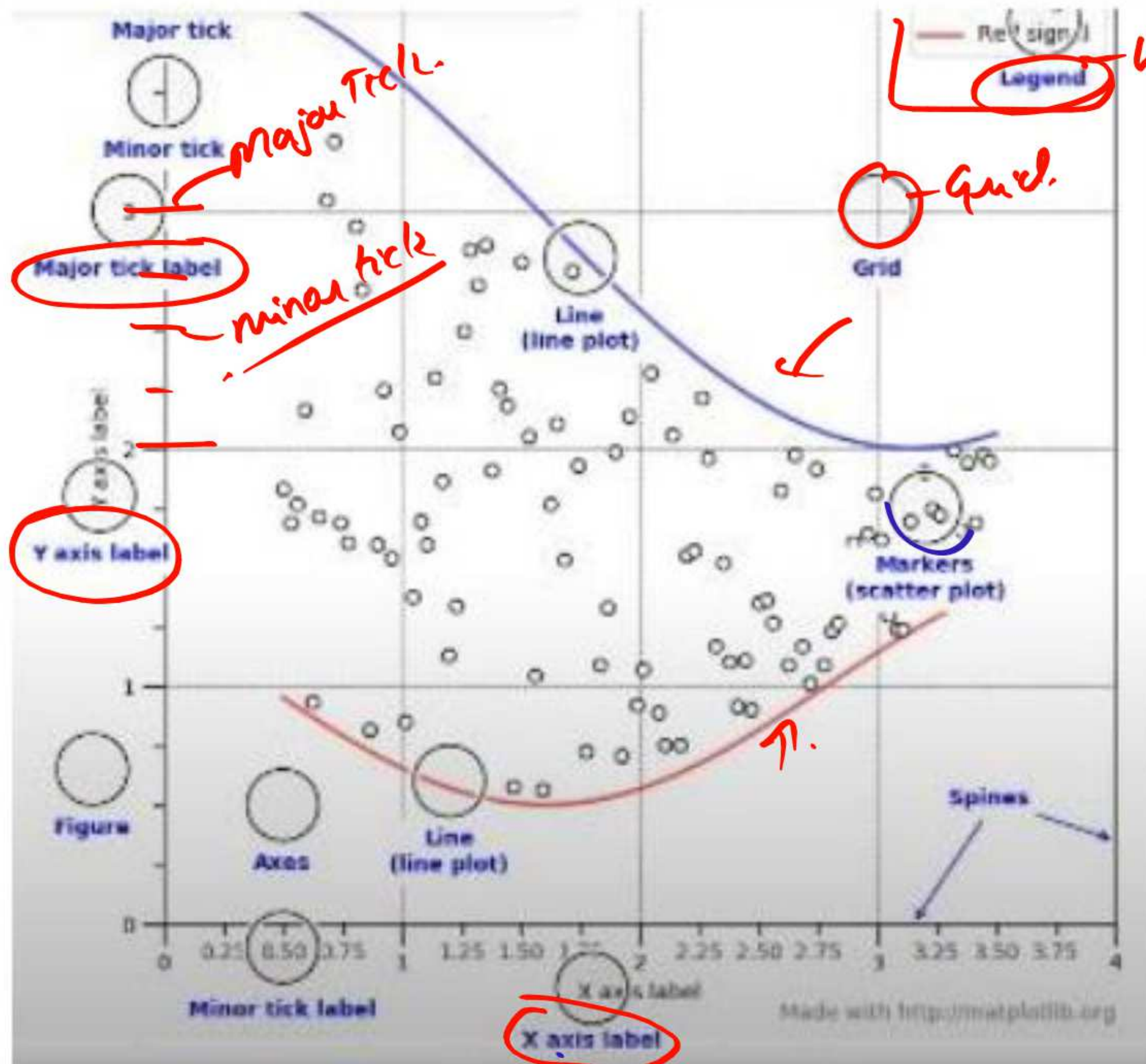
b = [4, 5, 6]

plt.plot($\begin{matrix} \vec{a} \\ \text{X} \\ \text{axis} \end{matrix}$, $\begin{matrix} \vec{b} \\ \text{Y} \\ \text{axis} \end{matrix}$)

plt.grid()

plt.show()






```
import matplotlib.pyplot as plt
```

```
x = [1, 2, 3, 4, 5]
```

```
y = [10, 20, 25, 30, 40]
```

```
plt.plot(x, y, color='blue', linestyle='--', marker='o', label='Growth')
```

```
plt.title("Line Plot Example")
```

```
plt.xlabel("Time")
```

```
plt.ylabel("Value")
```

```
plt.legend()
```

```
plt.grid(True)
```

```
plt.savefig("line_plot.png")
```

```
plt.show()
```

✓ Output:

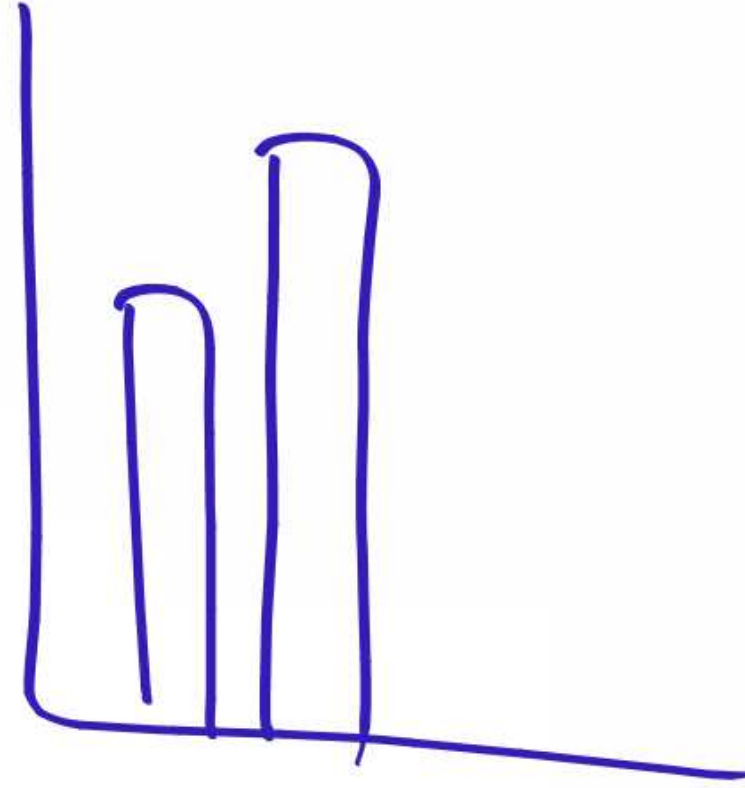
- A graph showing a dashed blue line with points marked and a legend.
- The plot is saved as 'line_plot.png' in the current directory.

```
categories = ['Math', 'Science', 'History']  
scores = [88, 92, 79]
```

```
plt.bar(categories, scores, color='green')  
plt.title("Student Scores")  
plt.xlabel("Subjects")  
plt.ylabel("Marks")  
plt.savefig("bar_graph.png")  
plt.show()
```

✓ Output:

- A green bar chart showing marks in 3 subjects.
- Saved as `bar_graph.png`.



Feature	Code
Title	<code>plt.title("Title Here")</code>
X-axis Label	<code>plt.xlabel("X Label")</code>
Y-axis Label	<code>plt.ylabel("Y Label")</code>
Grid	<code>plt.grid(True)</code>
Legend	<code>plt.legend()</code>
Save Plot	<code>plt.savefig("filename.png")</code>
Line Style	<code>linestyle='--' or '-.', etc.</code>
Marker Style	<code>marker='o' '*' 's' etc.</code>
Color	<code>color='red', 'green', '#FF5733' etc.</code>

How to Save Plots

python

```
plt.savefig("your_plot_name.png")
```

You can also use .jpg, .svg, or .pdf formats.


```
plt.plot(df["Salary"], color = "red", marker = "x", linestyle = ":", linewidth = "2")  
plt.grid()  
plt.show()
```



2 mins Summary



TEACHING
WALLAH

Topic

Data handling using Pandas

THANK YOU